



Mild synthesis of C6 + higher alcohols from aqueous ethanol upgrading over Ni@MB catalyst derived from melt-blown cloth

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ABSTRACT

Direct synthesis of alternative higher alcohols from aqueous bio-ethanol has received increasing attention because of its promising application in value-added fine chemicals and bio-jet fuels, yet the efficient carbon chain growth towards C6 + higher alcohols still facing great challenges. Herein, the polypropylene-based melt-blown cloth (MB) was adopted as novel precursor to fabricate the high-active Ni@MB catalysts for the direct mild upgrading of aqueous ethanol (50 wt%). Superior higher alcohols yield of 46.9 % with 74.1 % ethanol conversion was achieved at 170 °C over Ni@MB-1/20–500. The nitrogenous MB facilitated the formation of N-doped graphene structure that promoting the aldol-coupling and dehydrogenation capability of Ni@MB. For further enhancing the efficient carbon chain growth towards C6 + higher alcohols, a novel staged-reaction-cooling (SRC) strategy was developed and optimal C6 + higher alcohols selectivity up to 72.1 % with 83.2 % ethanol conversion was achieved. This work gains new insight to mild synthesis of C6 + higher alcohols from readily-accessible bio-ethanol.

1. Introduction

With the escalating energy and environmental issues, finding substitutes for petrochemical products is increasingly urgent. Biofuels and chemicals derived from renewable biomass are widely regarded as promising substitutes for traditional petrochemical products due to their numerous advantages, including low carbon emissions, abundant resources, and sustainable development (Tan and Tan, 2022; Ahmad et al., 2019). Bioethanol is a low-cost, widely available, and environmentally-friendly renewable energy source, showing its high degree of resource renewability and potential for circular utilization (Balat and Balat, 2009; Lynd et al., 1991). Higher alcohols like octanol have higher cetane numbers, greater energy density, lower hygroscopicity, and lower autoignition temperature compared to ethanol. As a result, the C6 + higher alcohols are considered as excellent diesel additives and jet fuel precursor (Nanthagopal et al., 2018; Damodharan et al., 2018; Luo et al., 2023; Luo et al., 2022; Zhang et al., 2023; Danfeng, n.d.). In addition, the C6 + higher alcohols can also find applications in plasticizers,

plasticizers, thickening agents, lubricants, detergents and other fields, offering tremendous economic value and market potential (Neto et al., 2018; Wang et al., 2017; Ambursa et al., 2020; Patel and Kumar, 2016).

Facing the turbulence in petroleum market, the alternative synthesis of higher alcohols instead of traditional olefin oxo-synthesis has gained increasing attention. The Guerbet condensation method is one of the most promising process to synthesis higher alcohols due to its efficient reaction pathway and the readily-accessible bio-ethanol resources (Liu et al., 2022; Wu et al., 2017; Lopez-Olmos et al., 2019; Hernández et al., 2016). The Guerbet coupling process typically involves three key steps. Firstly, ethanol is dehydrogenated to form acetaldehyde. Secondly, in the aldol condensation reaction, acetaldehyde undergoes a condensation reaction with another hydroxyl compound, resulting in the elimination of a water molecule and the generation of unsaturated aldehydes. Finally, unsaturated aldehydes are further hydrogenated to form higher alcohols. In this process, the first and third steps of the reaction require the participation of a metal active center, while the second step catalyzed by basic sites.

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Numerous catalysts have been developed for ethanol-coupling reactions (Table S1). Homogeneous catalysts, such as ruthenium, iridium, and other catalysts, exhibiting high butanol selectivity but still facing challenges in the catalyst separation (Wang et al., 2018; Wingad et al., 2015; Chakraborty et al., 2015). Bifunctional catalysts like hydrotalcite and hydroxyapatite usually require high reaction temperatures (above 300 °C) and result in severe side reactions, which leading to low carbon efficient towards higher alcohols. Noble metals (such as Ru, Rh, Pd, Au) combined with homogeneous bases can effectively accelerate the coupling reaction rate, but they are cost-prohibitive (Phung et al., 2014; Gao et al., 2018; Ni and Padilla, 2023; Sun et al., 2016). Currently, the primary products of ethanol coupling reactions are mainly low carbon alcohols such as butanol, while the selective synthesis towards higher alcohols like C6 + alcohols still facing challenges.

The porous carbon structures with electronic conductivity and hydrophobicity have gained attention for their hydrothermal stability and high surface area, which play crucial role in improving catalyst stability in aqueous environment and dispersing the active nano-particles (Huang et al., 2019; Wang et al., 2021; Liu et al., 2019). As a carbon-based polyolefins, the (polypropylene) PP has been widely used in plastic manufacture and account for huge plastic waste. Especially the medical protective equipment (MPE, like single-use face masks) enhances the generation of waste plastic since the outbreak of COVID-19 pandemic during past years. The melt-blown cloth (MB) is widely used for these MPE which is mainly composed of PP. The three-dimensional fiber network of MB brings numerous excellent properties as adsorbent material and carbon carrier, including a large specific surface area, oleophilic-hydrophobic characteristics, high adsorption efficiency and low cost (Yang et al., 2021; Wei et al., 2021; Nie et al., 2022; Wang et al., 2021). Additionally, as a catalyst carrier, MB may improve the dispersion of active metals and facilitate the catalytic efficiency and stability.

In this work, the melt-blown cloth (MB) was adopted as a novel carrier precursor to prepare the high-active Ni@MB catalysts for mild upgrading aqueous ethanol to C6 + higher alcohols. By screening the effect of Ni/MB ratio, carbonization temperature the MB was directly converted into graphene layer and optimized Ni@MB-1/20–500 catalyst was obtained. Superior catalytic performance of 46.9 % higher alcohol yield with 74.1 % ethanol conversion at 170 °C mild reaction temperature was achieved, and excellent hydrothermal stability and reusability of Ni@MB catalyst was examined. The nitrogenous MB precursor is crucial for reducing the graphitization degree and forming N-doped graphene defects, which promoting the dehydrogenation capability of Ni sites. In addition, optimal C6 + higher alcohols selectivity up to 72.1 % was firstly achieved by adopting a novel staged-reaction-cooling (SRC) strategy. This work gains new insight to mild synthesis of higher alcohols from readily-accessible aqueous ethanol and the waste-to-resource strategy.

2. Experimental section

2.1. Materials

All chemicals were commercially available and used as purchased

without further purification. Nickel (II) nitrate ($\text{Ni}(\text{NO}_3)_2 \cdot 6\text{H}_2\text{O}$, 99.8 %) was purchased from Macklin Biochemical Technology Co., Ltd. (China). Melt-blown cloth (MB) was obtained from Shixin Textile Company (China), with a basis weight ranging from 18 to 60 g/m² and average molecular weight between 40,000 and 90,000. Citric acid ($\text{C}_6\text{H}_8\text{O}_7 \cdot \text{H}_2\text{O}$, CA, with a purity specification of 99.5 %) was purchased from Aladdin Biochemical Technology Co., Ltd. (China).

2.2. Catalyst preparation

The Ni@MB catalysts were fabricated through a facile impregnation-carbonization strategy using melt-blown cloth (MB) and $\text{Ni}(\text{NO}_3)_2 \cdot 6\text{H}_2\text{O}$ as starting materials as shown in Fig. 1. Firstly, the $\text{Ni}(\text{NO}_3)_2 \cdot 6\text{H}_2\text{O}$ (1.6 g) was added into certain amount of ethanol to obtain a homogeneous nickel-ethanol solution. Then, the MB (unit weight of 25 g/m², 1.5 g) was fragmented and impregnated into the above nickel-ethanol solution with thoroughly stirring, followed by drying overnight at 80 °C in oven. Subsequently, the dried precursor was transferred to a tubular furnace and carbonized for 2 h at certain temperature with N₂ flow. Finally, the Ni@MB catalyst was obtained as black sample. The catalysts prepared with various conditions were denoted as Ni@MB-X-T, where X represented the molar ratio of Ni to MB (MB was calculated with PP basis, typical ratios of 1/5, 1/10, 1/15, 1/20, 1/25, 1/30, 1/35 and 1/50 were adopted), and T represented the carbonization temperature (typical temperatures of 400, 450, 500, 550, 600, 650 and 700 °C were adopted). For comparison, the Ni@CA catalyst was also prepared using same method except adopting citric acid (CA) as starting material instead of MB.

2.3. Catalyst characterization

The crystal states of obtained catalysts were analyzed by using X-ray diffractometer (MiniFlex 600, Rigaku, Ltd., Japan) equipped with a Cu K α radiation source and a wavelength (λ) of 1.5418 nm. The 2 θ scanning range was set from 5° to 80°, with a scanning rate of 10°/min. The microstructure and morphology were observed using scanning electron microscopy (SEM, SU8220, Hitachi) and transmission electron microscopy (TEM, FEI Thermo Talos F200S TEM). Prior to the characterization, the samples were thoroughly ground in an agate mortar. After 30 min of ultrasonic dispersion in anhydrous ethanol, the dispersed solution was dropped onto a carbon film supported on a copper grid for observation. The Brunauer-Emmett-Teller (BET) method was used to measure the pore characteristics and specific surface area of the materials at Micromeritics ASAP 2460. Prior to measurement, the samples were first degassed at 300 °C for 12 h. The Barrett-Joyner-Halenda (BJH) method was employed for further analysis of the data. Thermo Scientific K-Alpha X-ray Photoelectron Spectroscopy (XPS) analyzer was adopted to analyze the surface elemental composition and chemical states of the samples. Approximately 20 mg of sample was pressed into a pellet for testing, the acquired data was calibrated based on the binding energy of the C 1s peak at 284.8 eV. The CO₂ temperature-programmed desorption-mass spectrometry (CO₂-TPD-MS) experiment was conducted to analyze the desorption process of carbon dioxide on the catalyst.

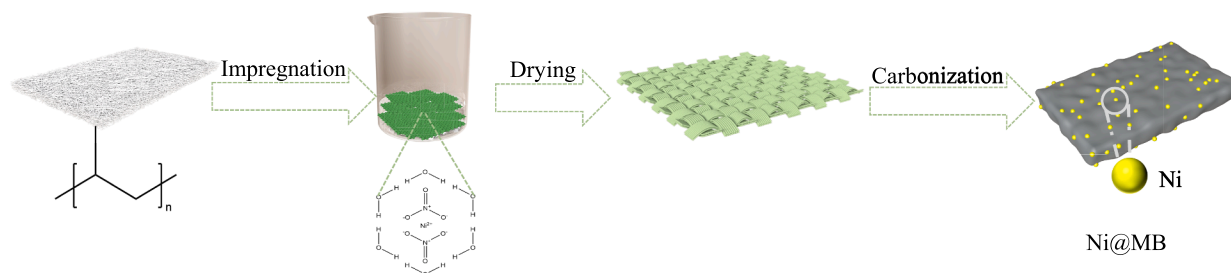


Fig. 1. Preparation process of the Ni@MB catalyst.

Raman spectrum was recorded using the WITec alpha300R system from Germany for carbon defect analysis. The content of Ni in the catalyst sample was analyzed using the Agilent 720 ES Inductively Coupled Plasma Atomic Emission Spectroscopy (ICP-AES) instrument from the United States. Additionally, the Vario EL cube GermanyElementar organic elemental analyzer was adopted to determine the content of C and N in catalysts.

2.4. Catalytic reaction and product analysis

The aqueous ethanol upgrading reaction was conducted in a 70 ml stainless steel magnetic stirring reactor (SSC70*4-Y2W6-316L-GY, Saifute Instrument Co., Ltd., Hefei, China). Typically, 10 g of 50 wt% aqueous ethanol solution, 0.87 g of sodium hydroxide and 0.3 g catalyst were added together to the autoclave and then sealed. The sodium hydroxide base catalyzed aldol condensation of aldehyde intermediates based on nucleophilic attack mechanism. Leakage checking was performed at 6 MPa prior reaction. After that, the reactor was flushed with hydrogen for 5 cycles to remove the air. Finally, the reactor was heated to 170 °C (unless otherwise stated) for the aqueous ethanol upgrading reaction, and the reaction lasted for 12 h with a stirring speed of 1000 rpm. After the reaction, the reactor was naturally cooled to room temperature. The gasses product was collected and analyzed on a Shimadzu GC 8860 gas chromatograph. The liquid product was centrifugated to obtain the aqueous and organic phase products respectively, then further analyzed based on the internal standard method on GC Shimadzu2010 Pro (HP-INNOWax column, 30 m × 0.32 mm × 0.50 μm, Japan).

For the reaction using staged-reaction-cooling (SRC) strategy, the reaction was performed same as the typical reaction process described above except using staged cooling reaction temperatures. Specifically, the reaction was firstly carried at a high temperature T_1 for t_1 duration, followed by cooling to a low temperature T_2 for t_2 duration, and the total reaction period kept as 12 h. Accordingly, the reaction sequence was denoted as SRC- T_1/t_1 - T_2/t_2 for discussion.

The catalytic performance was evaluated based on the ethanol conversion, higher alcohols yield and product selectivity according to the below equations:

$$\text{Conversion (Ethanol)} = 1 - \left(\frac{\text{Moles of EtOH after the reaction}}{\text{Moles of EtOH fed to the reaction}} \right) \times 100\%$$

$$\text{Yield (Higher alcohols)} = \frac{\text{Moles of carbon in the higher alcohols product}}{\text{Moles of carbon in ethanol fed to the reactor}} \times 100\%$$

$$\text{Selectivity} = \frac{\text{Moles of carbon in the target product}}{\text{Moles of carbon in all product}} \times 100\%$$

3. Results and discussion

3.1. Catalyst characterization

The catalyst support significantly influences the character of the catalyst, including the size of metal particles, their dispersion, stability, and surface electronic environment (Mahmoodi et al., 2019). Herein, the melt-blown cloth (MB) was used as the carbon support precursor to fabricate the Ni@MB catalyst. The composition and crystal structure of the catalysts were analyzed by XRD as shown in Fig. 2, where the distinct diffraction peaks at $2\theta = 44.5^\circ$, 51.7° , and 76.6° correspond to the (111), (200), and (220) planes of metallic Ni (JCPDS PDF #04-0850), respectively. As shown in Fig. 2a, the diffraction peaks of Ni strengthened as the Ni/MB ratio increased, along with narrower half-width. Combining with the ICP results (Table S2), it's found that the loading content of metallic Ni increases from 49.5 % to 66.2 % along with the Ni/MB ratio increases from 1/30 to 1/5. These results indicating the increase of Ni loading led to larger Ni particle size and reducing the Ni dispersion, which caused by the strong interactions between metal particles. Meanwhile, the carbonization temperature also has significant impact for the formation of Ni sites. As can be seen in Fig. 2b, the distinct peak corresponding to NiO is observed at 37° , and the diffraction peaks of Ni are quite weak when the carbonization temperature is below 450 °C. This indicates the pyrolysis temperature above 450 °C is necessary for the MB carbonization and in-situ reduction of Ni species. Nevertheless, the significantly enhanced Ni peaks above 600 °C suggesting the over-high carbonization temperature can led to severe metal agglomeration. For comparison, the Ni@CA catalyst was prepared as its XRD pattern shown in Fig. 2c, diffraction peaks corresponding to the Ni (111), (200) and (220) were also observed. Notably, the narrow FWHM of Ni@CA indicating larger Ni particle sizes compared to Ni@MB.

The N_2 adsorption and desorption results of the prepared Ni@MB catalysts are shown in Fig. 3. The catalysts exhibit typical Type IV isotherm within P/P_0 range of 0.4–1.0, indicating the characteristic of mesoporous network (Hongmanorom et al., 2021). The pore distribution analysis reveals the abundant mesopores with pore volume among 0.16 cm^3/g to 0.26 cm^3/g and BET areas among 151.1 m^2/g to 202.3 m^2/g (Table S2). Specifically, the high MB content as well as the carbonization temperature are beneficial for preparing Ni@MB catalyst with large specific surface areas, which is favorable to facilitate the accessibility of active sites.

The morphological characteristic of prepared catalysts was investigated by SEM and TEM as shown in Fig. 4 and Fig. S1. Clearly, the abundant multi-layered stacking structure of Ni@MB sample can be observed in Fig. 4a SEM image. Typically, the detailed TEM images further reveal the multiple carbon layers encapsulated metal particle structure of Ni@MB-1/20-500 catalyst (Fig. 4b–d), and the metal particles are well-dispersed with an average particle size of 5.63 nm

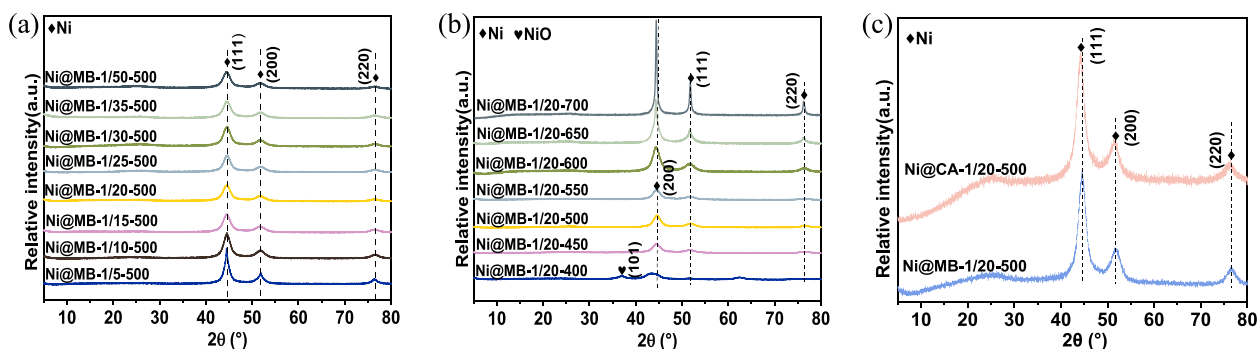


Fig. 2. The XRD diffraction pattern: (a) Ni@MB-X-500 catalysts, (b) Ni@MB-1/20-T catalysts and (c) Comparison between Ni@CA-1/20-500 and Ni@MB-1/20-500 catalysts prepared under same conditions.

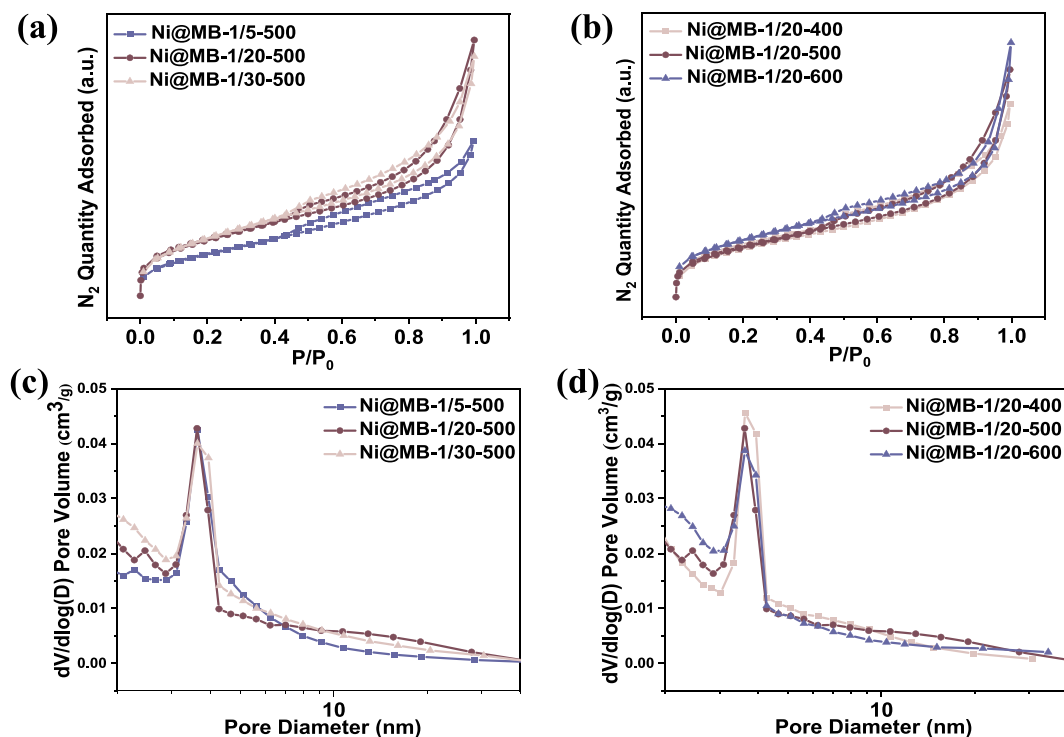


Fig. 3. (a-b) The N_2 adsorption and desorption isotherms and (c-d) Pore size distributions of Ni@MB-X-T catalysts.

(Fig. 4i). For Ni@MB prepared from different Ni/MB ratios and carbonization temperatures, the metal particle size increased with the increase of Ni/MB ratio (Fig. S1a–f) and metal aggregation occurred with elevated carbonization temperature (Fig. S1g–i), which is consistent with the XRD analysis mentioned. Additionally, lattice fringes with width of 0.203 nm are clearly observed, which is attributed to the diffraction of Ni (111) planes. It's worth noting that the energy-dispersive X-ray spectroscopy (EDS) scanning displayed in Fig. 4e showed a uniform distribution of nitrogen (N) element which is mainly distributed among the carbon layers, suggesting that N doping occurred during the carbonization process. The elemental analysis results further demonstrated the total N content of Ni@MB-1/20–500 is 1.63 % (Table S3), which is notably lower than the surface N content of 5.2 % obtained from XPS analysis. Generally, the detection depth of XPS is only the surface several nano-meters, thus this distinct difference between total and surface N contents further indicated the doping of N is mainly among the surface carbon layers (Kasera et al., 2021; Yamada et al., 2021; Murdoch et al., 2023). As the pure PP is nonnitrogenous, but the elemental analysis of MB confirmed its N content of 0.35 % (Table S3), thus these doped N should be derived from additives in MB. For example, antistatic agents, commonly used to prevent static electricity accumulation in meltblown fabrics, antibacterial agents used to improve the antibacterial properties of meltblown fabrics, and fiber stabilizers used to enhance the stability and light resistance of meltblown fabrics, usually containing nitrogen (Duan et al., 2022; Ren et al., 2020; Nafady et al., 2023; Ma et al., 2021). For comparison, the Ni@CA-1/20–500 catalyst prepared with CA as initial precursor showed larger average particle size of 8.20 nm and similar carbon encapsulated structure, which is consistent with the prior XRD analysis mentioned. Meanwhile, negligible N element was detected in the Ni@CA-1/20–500 catalyst (Table S3). As the N doping can enhance the interaction between metal and carbon layers, (Wang et al., 2023; Deng et al., 2022) thus the nitrogenous MB is favorable for the dispersion of Ni particles during carbonization process and it would contribute to high catalytic activity of Ni@MB catalyst.

The surface elemental composition and chemical states of the

prepared Ni@MB and Ni@CA catalysts were analyzed as shown in Fig. 5 and Fig. S2. It was evident that in Fig. S2a, the XPS survey spectra of the Ni@MB catalyst detected signals of Ni 2p, C 1 s, N 1 s, and O 1 s. In Fig. S2b, the C 1 s spectra indicated that 284.0 eV corresponds to C-C bond, 286.8 eV corresponds to C = O bond, 289.2 eV corresponds to O-C = O bond, (Wu et al., 2016) and 285.7 eV corresponds to C-N bond, demonstrating N doping in the carbon layer, consistent with the TEM results. The N 1 s spectra in Fig. 5a could be divided to multiple peaks located at 398.78 eV, 399.68 eV, 400.58 eV, and 401.38 eV, corresponding to pyridinic nitrogen, pyrrolic nitrogen, graphitic nitrogen, and oxidized nitrogen (Cao et al., 2017). The spectra of Ni 2p and O 1 s as shown in Fig. 5(b–c), revealed that Ni@MB catalyst prepared with carbonization temperature of 400 °C exhibited distinct NiO peaks, indicating the low carbonization temperature below 400 °C is unfavorable for the reduction of Ni sites, consistent with the XRD results. According to the surface elemental composition and chemical state of Ni@MB-1/20–500 and Ni@CA-1/20–500 catalysts prepared from different carbon precursors. Fig. 5d and f show that the Ni@CA-1/20–500 catalyst only detected signals of Ni 2p, C 1 s and O 1 s, without N 1 s, which is consistent with the previous elemental analysis results. Using citric acid (CA) as a precursor resulted in negligible N doping, while MB precursor enabled rich N doping.

Significantly, the nitrogen doping is conducive to the generation of carbon layer defects. In order to further analyze the carbon structure, Raman spectroscopy is used and the results are shown in Fig. 6a. The D peak (1350 cm^{-1}) represents defects in the carbon atomic lattice, while the G peak (1590 cm^{-1}) represents in-plane stretching vibrations of sp^2 -hybridized carbon atoms (Ammar et al., 2015). Typically, the I_D/I_G ratio is calculated based on the intensity of the D peak and G peak, and the high I_D/I_G ratio indicates abundant defects in the carbon atomic lattice (Dresselhaus et al., 2002). According to Fig. 6a, the I_D/I_G ratios for Ni@MB-1/20–500 and Ni@CA-1/20–500 are determined to be 0.999 and 0.832, respectively. Combined with the characterization results from XPS and EA, the reason for the reduction in graphitization of Ni@MB-1/20–500 can be attributed to the N doping, which disrupted the regular structure of graphite carbon. Since the N atom has one more

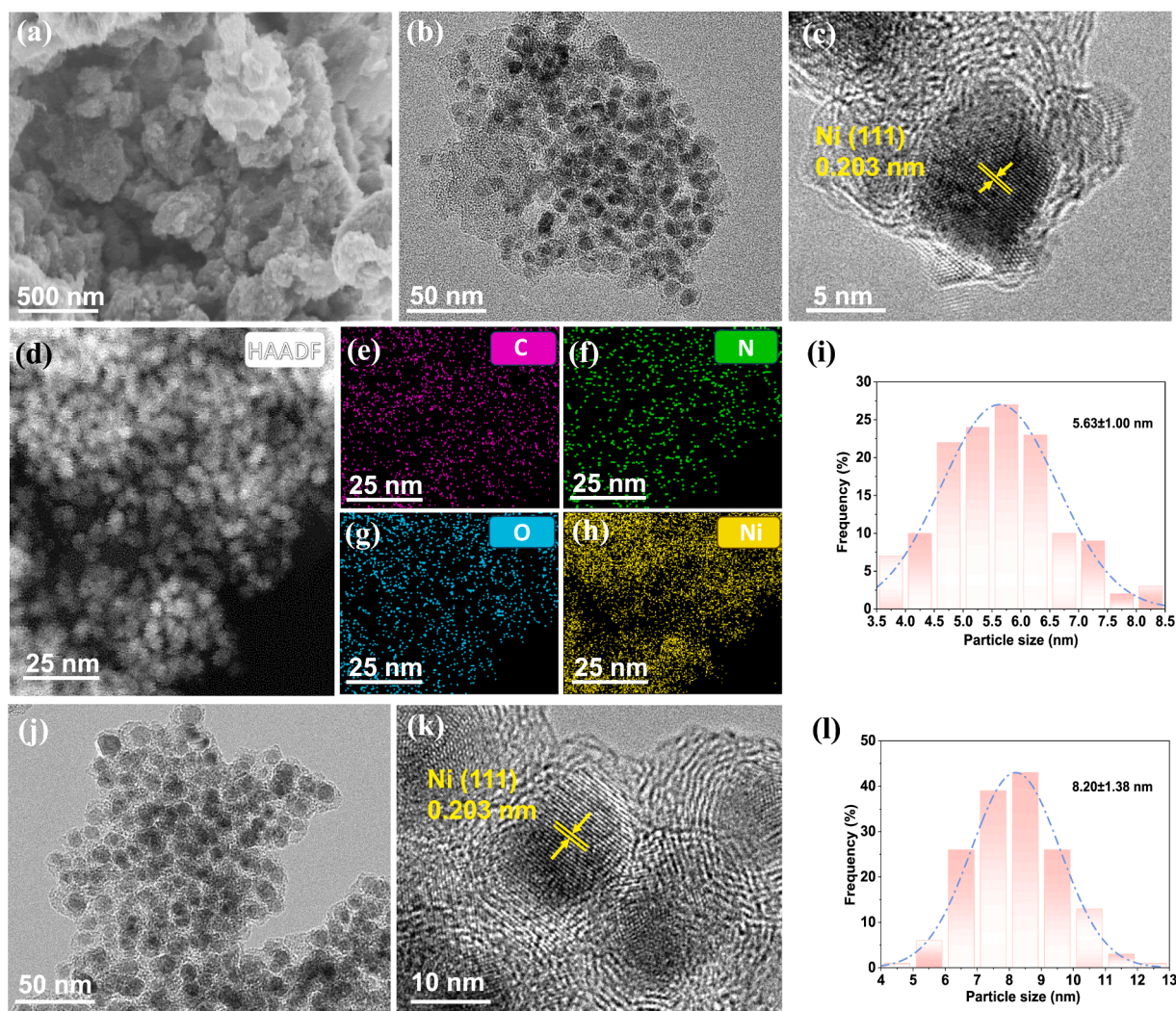


Fig. 4. (a) SEM image, (b) TEM image, (c) HRTEM image, (d) HAADF image, (e-h) EDS elemental map of C, N, O, and Ni, (i) particle size distribution for the Ni@MB-1/20-500 catalyst; (j) TEM image, (k) HRTEM image and (l) particle size distribution for the Ni@CA-1/20-500 catalyst.

valence electron than carbon atom, the doping of N increases the electron density of graphite layer, which affecting the electronic environment of neighboring Ni sites and leading to extraordinary catalytic performance (Jiang et al., 2018).

Specifically, the N doping in Ni@MB lead to additional pyridinic and pyrrolic N species as the XPS analysis indicated, and these species can introduce additional basic centers. The basic centers play a crucial role in ethanol coupling and dehydrogenation and aldol condensation (Feng et al., 2021). Therefore, the CO₂-TPD-MS analysis was further carried out to investigate the alkalinity changes of different catalysts. As shown in Fig. 6b and Table S4, the total alkalinity of Ni@MB-1/20-500 and Ni@CA-1/20-500 catalysts were 0.627 mmol/g and 0.198 mmol/g, respectively. Clearly, the Ni@MB showed enhanced basic amount, especially for the strong base site above 350 °C. According to XPS analysis, the enhancement of alkalinity in Ni@MB-1/20-500 catalyst is mainly attributed to the formation of pyridinic nitrogen (Zhang et al., 2023). In short, the MB precursor bring the N doping in the graphite layer and introduce additional basic sites, these would be benefit to the base-catalyzed aldol condensation C-C coupling process.

3.2. Catalytic upgrading of aqueous ethanol over Ni@MB

3.2.1. Effect of Ni/MB ratio on Ni@MB catalytic activity

Generally, the production produced from ethanol coupling included

liquid and gas product. The liquid product include normal alcohols (n-OH), isomeric alcohols (*iso*-OH) primarily composed of ethyl- and butyl-branches, intermediate aldehydes (-CHO) as byproducts, and over-hydrogenated hydrocarbons (HC). The gas product mainly consist of H₂ and C₁ gas products from C-C cleavage, such as CH₄, CO and CO₂. According to the catalytic performances of ethanol aqueous coupling shown in Fig. 7, the trends of ethanol conversion and higher alcohols yield varied with the change of Ni/MB ratio in Ni@MB-X-500 catalyst. When the Ni/MB ratio decreased from 1/5 to 1/20, the ethanol conversion increased from 64.6 % to 74.1 %, and the yield of higher alcohols also increased from 23.8 % to 46.9 %. More importantly, we found that the carbon distribution of higher alcohols shifted significantly towards C₆ + higher alcohols, with the highest selectivity reaching 66.7 %. In general, the carbon distribution of higher alcohol products from ethanol coupling usually shows decreasing trend with the increase of carbon number due to the step-growth cascade reaction process (Eagan et al., 2020). Here, the selectivity of C₆ hexanol reached 28.6 % compared with 29.9 % of C₄ butanol over the Ni@MB-1/20-500 catalyst as shown in Fig. 7b, which clearly broke the traditional cascading chain growth pattern of carbon distribution. This should be caused by the introduction of additional basic centers by N-doping, which enhances the aldehydes coupling (Ling et al., 2023; Juwen et al., 2022; Lin et al., 2022). Further decreased the Ni/MB ratio from 1/20 to 1/35, the conversion of ethanol and higher alcohols yield decreased accordingly.

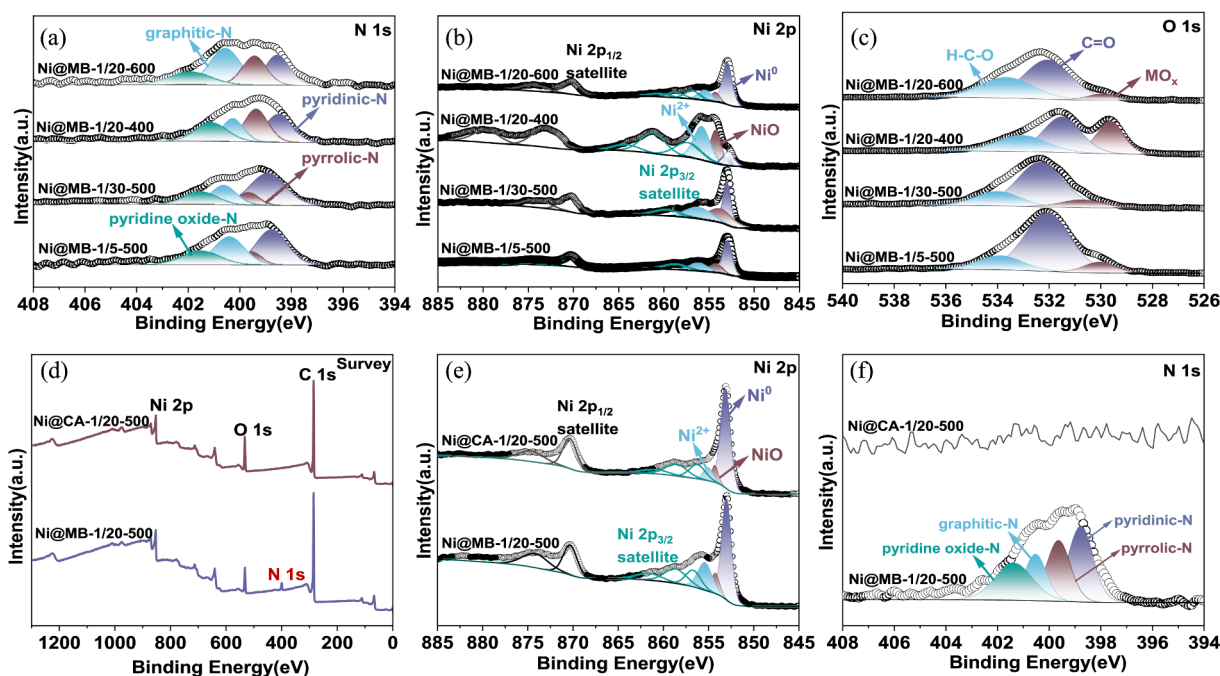


Fig. 5. XPS spectra of the Ni@MB-X-T catalyst: (a) N 1s spectra, (b) Ni 2p spectra and (c) O 1s spectra. XPS comparative spectra of the Ni@MB-1/20-500 catalyst and Ni@CA-1/20-500 catalyst: (d) Survey spectra, (e) Ni 2p spectra and (f) N 1s spectra.

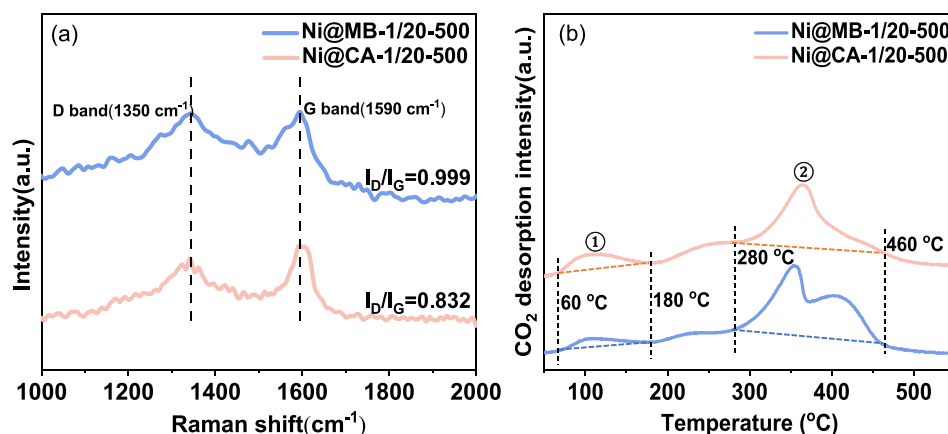


Fig. 6. (a) Raman spectra of the Ni@MB-1/20-500 catalyst and Ni@CA-1/20-500 catalyst, and (b) CO₂-TPD spectra of the Ni@MB-1/20-500 catalyst and Ni@CA-1/20-500 catalyst.

Combining with the prior XRD and TEM analysis, the high Ni/MB ratio led to larger particle size and metal aggregation, while the low Ni/MB ratio led to over-thick carbon layer, which both unfavorable for the accessibility of active sites (Zhu et al., 2022). Preferably, the catalyst with Ni/MB ratio of 1/20 exhibited best higher alcohols yield of 46.9 % with C₆ + higher alcohols selectivity of 66.7 %.

3.2.2. Effect of carbonization temperature on Ni@MB catalytic activity

The ethanol coupling catalytic activity using Ni@MB-1/20-T catalysts prepared at different carbonization temperatures were shown in Fig. 8. It can be observed that both ethanol conversion and higher alcohols yield enhanced as the catalyst carbonization temperature increased from 400 °C to 500 °C. But further increased the carbonization temperature to 700 °C led to significant decay of catalyst activity. Based on prior XRD and XPS analysis, the low carbonization temperature like 400 °C is inadequate for the formation of active-Ni and the over-high carbonization temperature led to severe metal agglomeration, which responsible for their declined catalytic activities. Optimally, the Ni@MB-

1/20-500 catalyst exhibited the best activity with ethanol conversion reached 74.1 %.

3.2.3. Effect of reaction temperature

The reaction temperature usually play crucial role for catalytic performance and the low reaction temperature is desired for its benefits in energy conservation and low reactor cost. Herein, the effect of reaction temperature on ethanol coupling to higher alcohols was investigated over the optimal Ni@MB-1/20-500 catalyst. As results shown in Fig. 9, the ethanol conversion increased significantly from 32.9 % to 93.3 % as the reaction temperature increased from 140 °C to 240 °C. However, the selectivity of undesired gas product also expanded with increased reaction temperature, which caused by enhanced C-C cleavage side reactions. At reaction temperatures of 240 °C, the gas by-product selectivity reached 50.7 %, which led to poor carbon atom efficiency toward target higher alcohols. Preferably, the Ni@MB-1/20-500 catalyst exhibited the optimal higher alcohols yield of 46.9 % and C₆ + higher alcohols selectivity of 66.7 % at 170 °C reaction temperature.

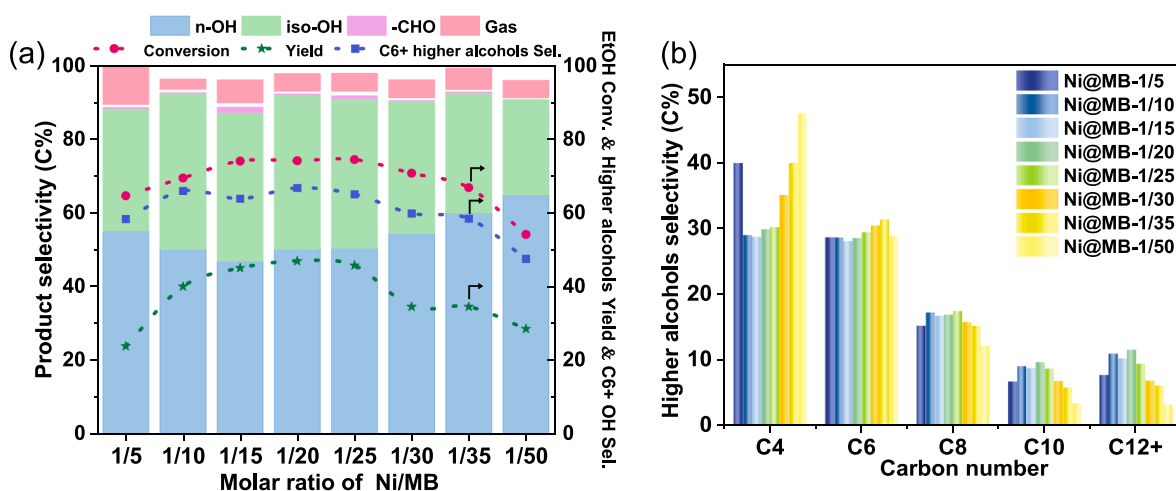


Fig. 7. Catalytic activity of ethanol to higher alcohols over Ni@MB catalysts prepared with different Ni/MB ratio: (a) ethanol conversion, higher alcohols yield, product selectivity and C6 + higher alcohols selectivity, and (b) carbon distribution of higher alcohol products.

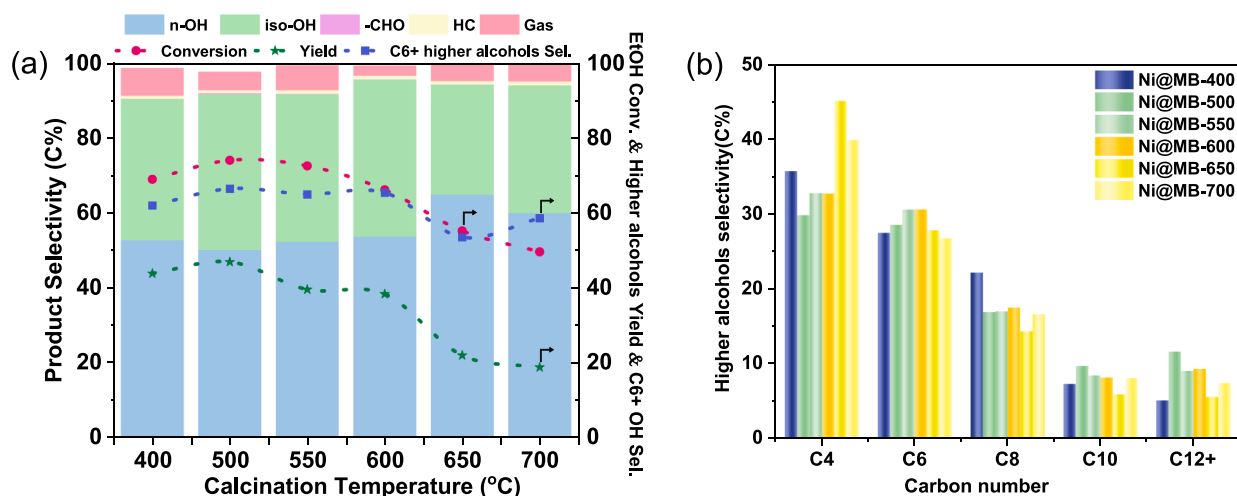


Fig. 8. During the preparation of the Ni@MB-1/20-T catalysts, the influence of different carbonization temperatures on the conversion of ethanol to higher alcohols: (a) ethanol conversion, higher alcohols yield, product selectivity and C6 + higher alcohols selectivity, and (b) carbon distribution of higher alcohol products.

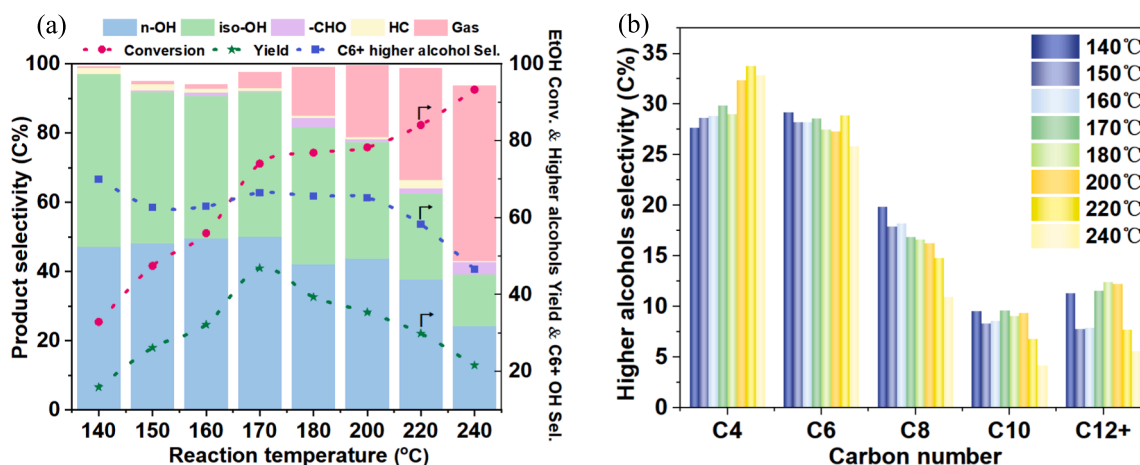


Fig. 9. The influence of reaction temperature on the conversion of ethanol to higher alcohols: (a) ethanol conversion, higher alcohols yield, product selectivity and C6 + higher alcohols selectivity, and (b) carbon distribution of higher alcohol products.

Meanwhile, compared to the previously reported catalyst (Table S1), the results clearly demonstrate that the Ni@MB-1/20–500 catalyst exhibits significantly superior ethanol conversion rate and higher alcohols yield in the ethanol coupling with mild reaction temperature.

3.3. Superior ethanol coupling activity of Ni@MB catalyst

To further demonstrate the beneficial effect of the nitrogen-containing MB precursor on catalyst activity, we prepared the Ni@CA-1/20 catalyst using nitrogen-free citric acid (CA) as precursor under the same conditions and the activity comparison is shown in Fig. 10. For the Ni@CA-1/20–500 catalyst, the ethanol conversion reached 67.7 %, the high carbon alcohol yield reached 34.3 % with C6 + higher alcohols selectivity of 61.3 %. In comparison, the Ni@MB-1/20–500 catalyst showed clear activity advantage towards higher alcohols production. An obvious observation was that the Ni@MB-1/20 catalyst led quite low selectivity towards gas by-products (4.7 %), indicating that the cleavage of C-C bonds was suppressed. This is related to the electronic regulation of Ni sites by N doping according to previous researches (Juwen et al., 2022). According to the carbon number distribution in Fig. 10b, the Ni@CA-1/20–500 catalyst contributed to a typical decreasing distribution of higher alcohols with the carbon number increasing. However, the Ni@MB-1/20–500 catalyst broke such distribution, which produced more C6 + alcohols (66.7 %) and the C6 hexanol reached similar selectivity with C4 butanol. This indicates the further conversion of butanol was promoted over Ni@MB.

Generally, the coupling of ethanol involves multiple steps of alcohol dehydrogenation, aldehyde condensation and olefinic alcohol hydrogenation. The metal site plays a crucial role in the hydrogen transfer reaction, while the aldol condensation process mainly catalyzed by alkali (Pang et al., 2020). To further investigate the superior activity of Ni@MB, the catalytic dehydrogenation of butanol was conducted over Ni@MB-1/20–500 and Ni@CA-1/20–500 catalysts under alkali-free conditions. As shown in Fig. 11, the butanol conversion reached 17.6 % over Ni@MB catalyst, which is clearly higher than that over Ni@CA. Meanwhile, the selectivity of aldehyde was 26.3 % with Ni@MB catalyst, as well as the selectivity of C8 + higher alcohols was 72.3 %. These indicate that the Ni@MB catalyst effectively promoted the butanol dehydrogenation and the butyraldehyde aldol condensation reaction without the help from external base. In contrast, the selectivity of aldehyde was only 64.1 % over Ni@CA catalyst, and non C8 + higher alcohol was detected. The results above indicate that the Ni@MB catalyst has strong butanol dehydrogenation capability, which is crucial for the further coupling of butanol intermediate during the ethanol to higher alcohols. Meanwhile, the additional basic sites in graphite layers brought by N doping in Ni@MB promoted the rapid conversion of the

aldehyde intermediate, which is consistent with the previous CO₂-TPD experimental results.

Combining the results of butanol dehydrogenation experiment and prior analysis, the possible reaction mechanism pathway was proposed as shown in Fig. 12. The Ni@MB catalyst prepared using N-containing MB possess abundant N-doped graphite layer, which introduced additional basic sites and enhanced the charge transfer between Ni and defect graphite. Consequently, these basic sites promoted the rapid conversion of aldehyde intermediates, inhibiting the deep dehydrogenation and C-C cleavage of aldehydes. Meanwhile, the further conversion of higher alcohols intermediate product such as C4 butanol was promoted, thus accelerate the step-growth ethanol coupling process to produce C6 + higher alcohols (Chen et al., 2022).

3.4. Recycle stability of Ni@MB catalyst

The hydrothermal reaction environment generally harsh for catalyst stability. To further evaluate the reusability and stability of Ni@MB catalyst, the recycle experiment were performed on the spent Ni@MB-1/20–500 catalyst. The spent catalyst was directly recovered by convenient centrifugation, then washing by ethanol and water to remove surface residues. The recovered spent catalyst was directly used for subsequent cycling experiment. The catalytic stability and characterization of Ni@MB catalyst are shown in Fig. 13. It's clear that the ethanol conversion and higher alcohols yield kept similar in 5th recycles, and the selectivity of gas by-products kept below 7 % throughout the recycle period. Meanwhile, it's observed that the crystal structure and surface chemical state of the catalyst remained stable compared to fresh catalyst during the recycle process (Fig. 13b). These results indicate that the Ni@MB-1/20–500 catalyst exhibited excellent stability and reusability, which can be attributed to graphene layer encapsulated Ni core-shell structure. The water-stable graphite layers provide strong protection for the metal cores in hydrothermal reaction environment.

3.5. Staged-reaction-cooling (SRC) strategy for enhancing higher alcohols synthesis

The traditional ethanol coupling reactions require high reaction temperature to overcome the alcohol dehydrogenation barrier, especially for the further dehydrogenation of higher alcohol intermediates (Liu et al., 2021). Actually, the aldol condensation process as the key step for C-C coupling during ethanol upgrading, does not need high reaction temperature to proceed, it can carry out even at room temperature with homogeneous base as catalyst. On the contrary, the high temperature can facilitate the C-C cleavage side reactions of generated aldehyde intermediates, which results in producing lots of gas by-

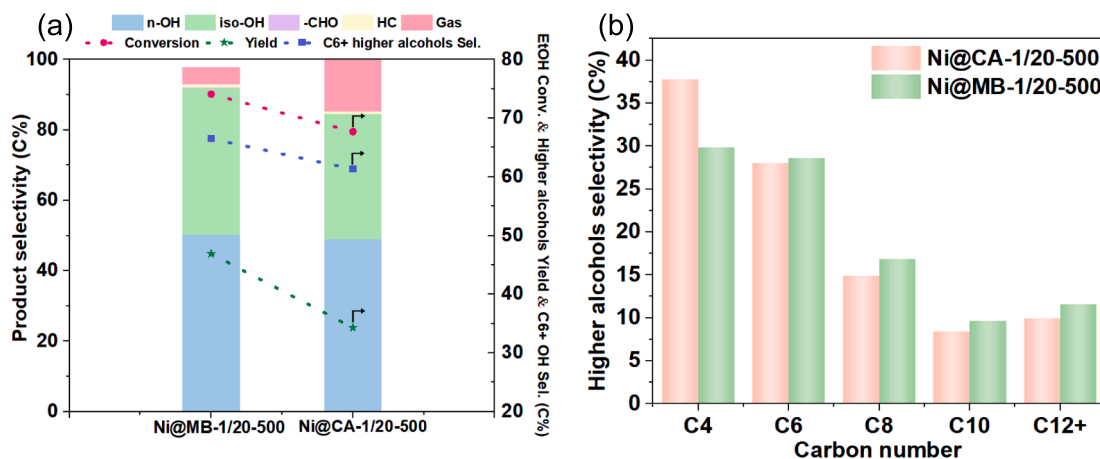


Fig. 10. Catalytic performance of aqueous ethanol coupling over Ni@MB-1/20–500 and Ni@CA-1/20–500 catalysts: (a) ethanol conversion, higher alcohols yield and product selectivity, (b) Carbon distribution of higher alcohol products. (Reaction conditions: Catalyst 0.3 g, NaOH 0.87 g, H₂O 5.0 g, EtOH 5.0 g, 170 °C for 12 h).

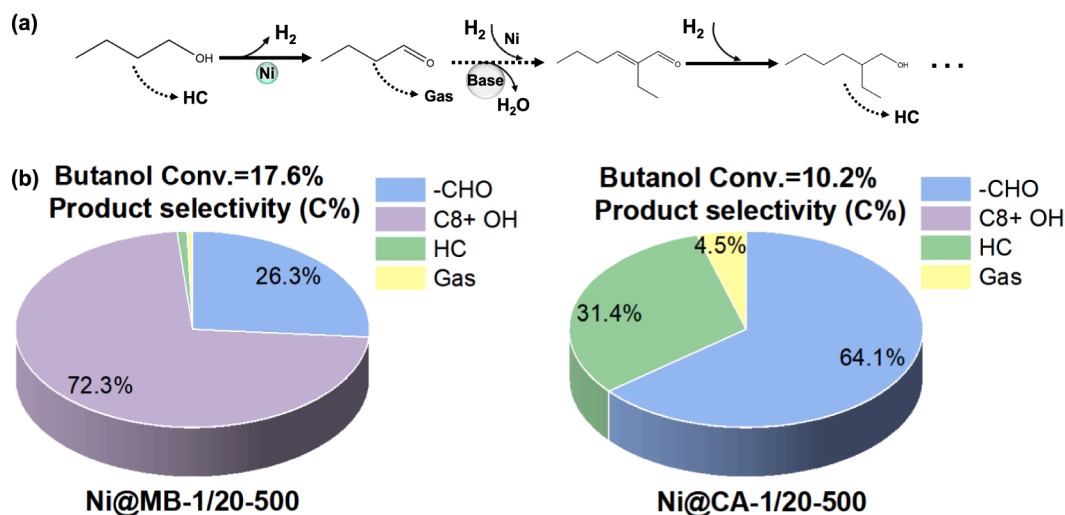


Fig. 11. Catalytic dehydrogenation of butanol on Ni@MB-1/20-500 and Ni@CA-1/20-500 catalysts without external base: (a) schematic possible reaction mechanism pathways, (b) catalytic performance of Ni@MB-1/20-500 catalyst and (c) catalytic performance of Ni@CA-1/20-500 Catalyst. (Reaction conditions: Catalyst 0.3 g, H₂O 5.0 g, Butanol 5.0 g, 170 °C, 12 h).

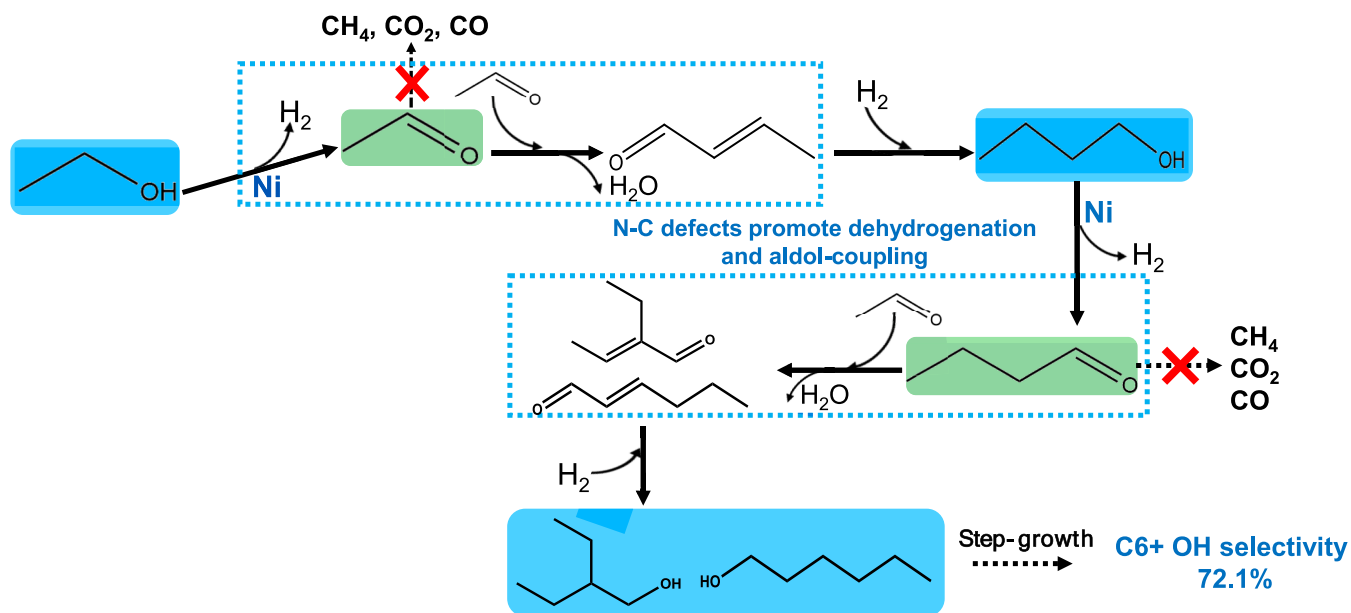


Fig. 12. Schematic possible reaction mechanism pathways in aqueous ethanol upgrading over Ni@MB catalyst.

products, thereby reducing the carbon atom conversion efficiency towards desired higher alcohols. Based on this, changing the reaction temperature to meet the different requirements of dehydrogenation and aldol condensation sequential steps seems an effective strategy to obtain the enhanced production of higher alcohols.

According to the above hypothesis, we proposed a staged-reaction-cooling (SRC) strategy, which involves a high-temperature reaction stage followed by a low-temperature reaction stage. Specifically, the reaction temperature was firstly set to high-temperature stage of 170–220 °C for 6 h, and then decreased to low-temperature stage of 140–170 °C for another 6 h. According to the results shown in Table S5 and Fig. S3, the ethanol conversion reached 81.1 %, and the yield of higher alcohols reached 40.9 % with C6 + higher alcohols selectivity of 72.6 % for the SRC-200/6–150/6 process, which are significantly higher than that of general 200/12 and 150/12 reaction process. Besides, it's worth noting that the selectivity of C6 alcohols was clearly higher than C4 alcohols (Fig. S3), which broke the restriction of progressive decline

selectivity from butanol to the long-chain alcohols in the classic step-growth ethanol coupling model.

To further explore the effect of different SRC conditions, a series control experiments were performed as shown in Table 1. For the SRC-200/6-T₂/6 reactions, with the increase of T₂ temperature (low-temperature stage) from 140 °C to 170 °C, the higher alcohols yield and C6 + alcohols selectivity first increased then declined, which indicates the high temperature above 150 °C promoted the C-C cleavage. As for the SRC-T₁/6–150/6 reactions, the increase T₁ temperature from 170 °C to 220 °C clearly enhanced ethanol conversion from 64.4 % to 81.4 %. However, the elevated temperature also enhanced C-C cleavage reactions, which led to the continuous decrease of higher alcohols yield from 44.1 % to 25.1 %. In the meantime, the C6 + alcohols selectivity increased from 64.8 % to 72.6 % with T₁ increased from 170 °C to 200 °C, which should be attributed to the promoted generation of higher aldehydes from higher alcohols dehydrogenation at high temperature. Based on these results, the T₁ of 200 °C and T₂ of 150 °C were preferred

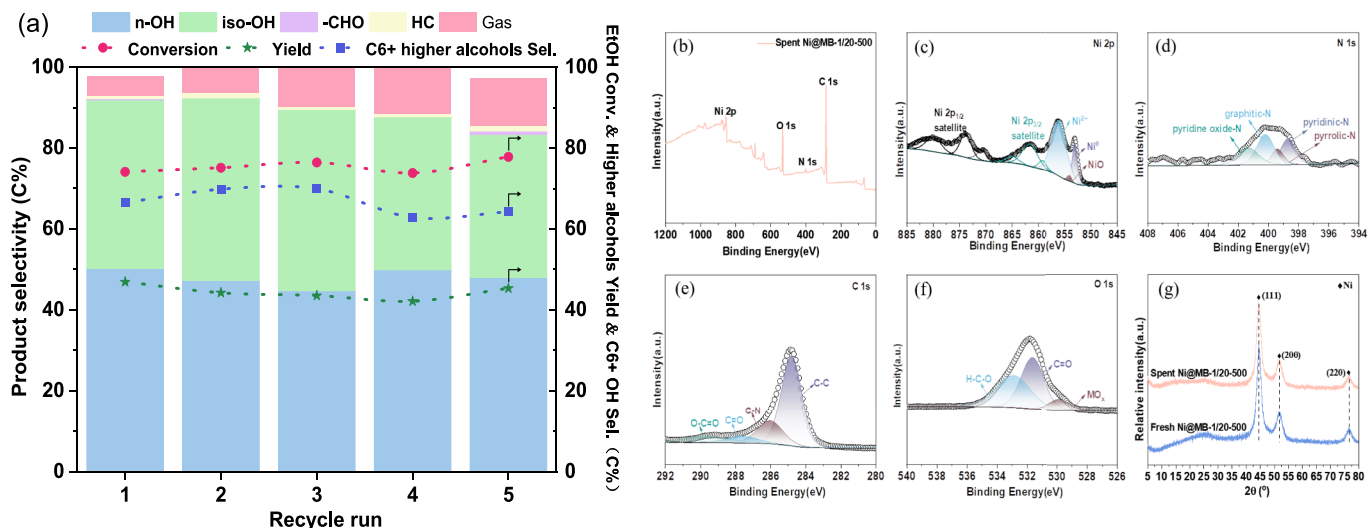


Fig. 13. (a) The cycling performance of the Ni@MB-1/20-500 catalyst (The catalyst used for the first run was 0.301 g, the mass of remaining catalyst for 2th cycle was 0.285 g, 3th cycle was 0.268 g, 4th cycle was 0.248 g and the 5th cycle was 0.226 g), the XPS spectra of spent Ni@MB-1/20-500 catalyst: (b) Survey spectra, (c) Ni 2p spectra, (d) N 1s spectra, (e) C 1s spectra, (f) O 1s spectra, and (g) XRD spectra.

Table 1

Ethanol upgrading performance over Ni@MB-1/20-500 catalyst with SRC strategy.

Strategy	T ₁ / °C	t ₁ / h	T ₂ / °C	t ₂ / h	EtOH Conv./C %	-OH Yield/C %	Selectivity (C6 + OH)/C%
SRC	200	6	140	6	77.5	40.5	67.0
	200	6	150	6	81.1	40.9	72.6
	200	6	170	6	76.9	38.1	65.5
	170	6	150	6	64.4	44.1	64.8
	220	6	150	6	81.4	25.1	62.6
	200	4	150	8	79.9	39.0	70.2
	200	8	150	4	83.2	42.5	72.1
	200	10	150	2	82.0	34.4	68.8
	200	10	150	2	82.0	34.4	68.8
General	T/ °C	t/h	–	–	EtOH Conv./C %	-OH Yield/C %	Selectivity (C6 + OH)/C%
	200	12	–	–	78.3	35.4	65.2
	170	12	–	–	74.1	46.9	66.5
	150	12	–	–	47.5	26.2	62.7

Reaction conditions: 0.3 g Ni@MB-1/20-500 catalyst, NaOH 0.87 g, H₂O 5.0 g, EtOH 5.0 g.

for the selective synthesis of C6 + higher alcohols. In addition to SRC temperatures, the different duration of SRC stages also affect the catalytic performance. As the t₁ duration (high-temperature stage) increased from 4 h to 10 h, the ethanol conversion kept at around 80 %, while the higher alcohols yield achieved its highest of 42.5 % at 8 h and plunged to 34.4 % as t₁ prolonged from 8 h to 10 h. This result indicates the over-long high-temperature stage only failed to facilitate the ethanol conversion, but also enhanced the side reactions. Given the above, the SRC strategy is quite effective for enhancing higher alcohols synthesis, and it's crucial to strike a balance between enhancing the higher alcohols dehydrogenation and suppressing the C-C cleavage of aldehydes during adopting the SRC strategy for the controlled synthesis of C6 + higher alcohols. Optimally, the SRC-200/8-150/4 reaction achieved ethanol conversion of 83.2 % and higher alcohols yield of 42.5 % with C6 + alcohols selectivity of 72.1 %.

4. Conclusion

In summary, the mild synthesis of C6 + higher alcohols from direct aqueous ethanol upgrading were achieved over the high-active Ni@MB

catalyst derived from melt-blown cloth (MB). The nitrogenous MB facilitated the formation of graphite defects and forming N-doped graphene layers which bring additional basic sites, thus promoting the aldol-coupling and dehydrogenation capability of Ni@MB. Benefiting from this, superior higher alcohols yield of 46.9 % with 74.1 % ethanol conversion was achieved at 170 °C mild reaction temperature over optimal Ni@MB-1/20-500 catalyst. Excellent hydrothermal stability and reusability of Ni@MB catalyst was demonstrated. Furthermore, a novel staged-reaction-cooling (SRC) strategy was proposed for enhanced synthesis of C6 + higher alcohols from aqueous ethanol coupling, the SRC reaction conditions were explored and optimal C6 + higher alcohols selectivity up to 72.1 % was firstly achieved. This work gains new insight to mild synthesis of alternative higher alcohols from readily-accessible aqueous bio-ethanol and the waste-to-resource strategy.

CRedit authorship contribution statement

Qian Zhang: Writing – review & editing, Validation, Supervision, Methodology, Data curation. **Yanlan Cen:** Writing – original draft, Visualization, Software, Project administration, Methodology, Investigation, Formal analysis, Data curation. **Bo Chen:** Conceptualization. **Jiajin Li:** Methodology. **Xinqi Zheng:** Visualization. **Xiaoping Wu:** Project administration. **Songbai Qiu:** Formal analysis. **Tiejun Wang:** Resources.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The data that has been used is confidential.

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Appendix A. Supplementary data

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